

Plotka Vision Statement

- "Planning Trips Made Easy!"
- <https://github.com/ArnavPawar/travelChatbot>



Executive Summary

Elevator Statement:

Nobody knows the perfect travel plan for any destination, but Plotka can provide information on all your travel needs/wants but also cater to your specific needs. We are an online Travel chatbot that takes input from you and uses it to create your dream trip. We provide details on flights, hotels, activities, and restaurants, as well as pricing for your trip.

Problem and Proposed Solution:

When someone is planning a vacation, the first thing that comes to mind is how much it will cost. Additionally, the lack of familiarity with the area can hinder the ability to find quality dining, living quarters, or knowing what to pack. Our product solves these problems by allowing users to input their parameters for a vacation, such as location and budget, and gives them quality recommendations.

These recommendations would come from reviews submitted online, the type of food served, and proximity to the place that the individual would be staying/visiting that day. In addition, We will apply for the API from Google Maps and then get information about the different locations. After that, We will design a standard and calculation, like machine learning, to evaluate each path API provides. Last not but not least, we will take advantage of chatgpt API to help us give customers the most appropriate recommendations according to their requirements.

Overview of the Development Process:

Development will be focused on the chatbot's ability to provide accurate and quality recommendations based on user input. Development will build on itself each week during our Sprints. This will include research, code implementation, and documentation. We plan to have a working model each week starting at the end of Sprint 3 or 4.

Overview of the Schedule:

Sprints will begin on Sunday at 2pm and end the following Friday at 11:59PM. We will hold weekly meetings to review our task for the weekly sprint and update our status report. Small meetings will be held with TA Mav Modi every Wednesday at 10:30AM.

Overview of Risks:

- The large size of backend projects will lead to the failure of online upload of api. Our application will not run if this occurs and the chatbot will not function. In addition, Too many pl-ins in the large project can make it difficult to find problems.
 - Possible Mitigation: We can split a large website into multiple small HTML formats and then connect them, such as Springboot.
- Slow speed of information generation will lead to users attempting to find another application due to frustration.
 - Possible Mitigation: Create faster generation algorithm OR faster way of pulling information.
- The high failure rate of map generation and tour guidance will reduce customer comfort. This is because, in our interface, the number of simultaneous responses and daily responses is limited
 - Possible Mitigation: Use Multiple response interfaces occur simultaneously to generate multiple guidance and map as sources.
- Failure to update data in real time will result in an incomplete application result.
 - Possible Mitigation: Test repeatedly and use multiple sources so that if one source were to not send out data we would still have a pool to sample from.
- May fail to quickly obtain the information required by users on the Internet
 - Build an effective database
 - Temporarily limit the range of service areas

Feature List

1. Questionnaire: A list of questions given to the user that would help the AI plan a more accurate trip. Questions will include: Budget, departure location, destination, number of people and ages, interests/hobbies, food interests, and flight class preference/ airline.
2. Day Trip Planner: Based on the questions answered by the user the AI would plan a trip and give a day-by-day trip that the user could potentially go on.
3. Packing List: Give a list of items the user should pack based on the location, activities they plan on doing, and temperature.
4. Restaurant Recommendations: 10 restaurant recommendations appear for the user based on budget and food preferences.
5. Hotel Recommendations: 10 hotel recommendations appear for the user based on budget and location preferences.
6. Activities: Recommends Activities in the desired destination that have good reviews. This can be physical activities or something as simple as visiting historic landmarks
7. Estimated Cost: Budget and estimated cost for the trip. Should give a close to accurate estimation of the trip based on the trip planner.
8. Website Categories: Have separate categories on-site to store all the different data mentioned above. Categories will include travel details(how to get from A-B and what

location A and B is), a packing list(section where all items are), an areas of interest section, a restaurant recommendation section, and a hotel recommendation section.

9. Customizable trip (post planning): Have a separate chat feature to customize trips even after a planned trip is provided.
10. Anonymous user reviews of website: shown the ratings by other users attract the website visitors to utilize the implementation.
11. Flight recommendation: 10 flight recommendations appear for the user based on the trip schedule.
12. Filter: A filter that provides different input based on user preferences, for example a huge budget input would result in first class seats and 5 star hotels.
13. User Match: Match users with people who have the same travel targets (same destination same interest close budget ...)
14. User login: includes username and password
15. Backstage service: Helping service for customers, including the operating guidance for the platforms.

Business Case

Niche:

- Aimed at tourists and people who want to travel on vacation with detailed plans.
- Used by people who want to have a more simplified vacation planning process and avoid the hassle of searching through several different sites for travel recommendations.

Competition:

The current navigation tools:

- Trivago
 - <https://www.trivago.com/>
- Expedia
 - <https://www.expedia.com/>
- TripAdvisor
 - <https://www.tripadvisor.com/>
- Google Maps
 - <https://www.google.com/maps>

Booking.com:

Currently is planning on creating a trip planner ai, that may be able to replicate some of the features we plan on implementing such as showing recommendations for activities, restaurants and hotels. With the surge of ai technology due to chatGPT most travel sites are eager to create an ai tool that helps with booking, but the market is empty and there is no dominant product out yet in the market.

Besides, this company owns a large amount of customer data and unique machine learning technology. A large amount of data means that they can improve their design formulas frequently until their customers satisfy their services. However, the lack of data means that we can only take advantage of some FOSS from github to improve the performance of our project.

- <https://globalnews.booking.com/bookingcom-launches-new-ai-trip-planner-to-enhance-travel-planning-experience/>

Our web app would allow users to build a trip from start to finish. All of the softwares or web applications mentioned above only provide either flights, or hotel locations, or places to go. However, none can do all of the things mentioned and more. Customers still need to spend a long time designing their plan by themselves. Our application will provide several fully planned trips with close to all information a user would need.

Value for the effort:

- Conduct a large number of investigations to improve the comfort of user travel
- Reduces time for planning a vacation online
- More readable UI
- We will have features which our competitors do not, such as post planning editing and user matching.
- We will try our best to collect data from customers, then we can add these data into our machine learning technology.

Risk of Product Success:

- The network security cannot be ensured.
- It is difficult to get large numbers of user clicks in a short time.
- It takes a lot of money for advertising.
- Hard to get people to trust AI to plan a trip for them.

Mitigating Risk

- We will have the frontend to be appealing to the user, have it be simple and easy to use
- Will make sure the site is fully tested with numerous user story examples.
- Ensure advertisement shows proper use of site with a thorough output

Stakeholders

Development Team:

The team consists of six RPI undergraduate students majoring in Computer Science. The team's role is to design the Plotka application on the website. The designers are invested in the program's success because they can achieve a high grade in the course for their project success and improve their design skills, and the possibility of continued development after the class concludes.

Development Team:

- Arnav Pawar
- Alex Litchfield
- Anson Decker
- Yuecheng Liu
- Yihan Shao
- Ziqun Wang

Prof. John Sturman and TA's:

Professor John Sturman and the TA's/mentors have roles as stakeholders. As the esteemed instructor of this software design and documentation course, he not only imparts valuable knowledge and guidance but also sets academic standards and evaluates the team's progress. He motivates the students to succeed in their design projects and guides the team in the right direction throughout the project's lifecycle. His involvement reinforces the practical relevance of the project and fosters a real-world learning experience that prepares the team members for future challenges in software design.

Professor:

- John Sturman

TAs:

- M. Shahid 'Mav' Modi

Mentors:

- Dong Lin
- Nathen Hernandez

***Note: This Project Schedule may change over time**

Assignment(s)	Projected Start Date:	Projected Due Date(Sprint #)
Get vision statement, Contributions, Status report, and organize schedule	N/A	Sept 22nd(S1)
Design the api Design and draw a diagram for what the web page looks like. Begin working on the site. Begin working on displaying basic ai api output	Sept 24th	Sept 29th(S2)
Parse in api data and format it to display on site Finalize our basic site. Features should include (Trip Plan, Packing list, restaurant recommendations, and hotel recommendations). Work on Sprint 4 deliverable	Oct 1st	Oct 6th(S3)
Fix Bugs finalize all features previously mentioned. Finish the deployment diagram SPRINT 4 DELIVERABLES DUE Renew the project schedule based on the finished work Begin working on review section for site	Oct 8th	Oct 13th(S4)
Work on backend and storing user login with planned trip/trips Make frontend more user friendly Implement image display and links for certain things like food spots, recommended travel spots, ect. Begin work on the following features: Have a basic user login functionality that stores the user's trip. Have anonymous review section	Oct 15th	Oct 20th(S5)
Get everything running smoothly, run tests involving preset inputs and what we expect for an output. Each test will be a certain user story example. Work on Interim Release	Oct 22nd	Oct 27th(S6)
Interim Release Deliverables due: Features: Estimated Cost, Activities	Oct 29th	Nov 3rd(S7)

Test the features listed in the Interim Release and work on image display for the site. Make sure images are ready for beta release. Flight recommendations and new features we may come up with.	Nov 5th	Nov 10th(S8)
Beta Release Deliverables Due: All features should be completed smoothly.	Nov 12th	Nov 17th(S9)
FINAL RELEASE DUE	Nov 19th	Nov 24th(S10)

Features to be completed by Interim Release:

- Questionnaire
- Day Trip Planner
- Packing List
- Restaurant Recommendations
- Hotel Recommendations
- Basic Login
- Anonymous Reviews
- Activities
- Estimated Cost
- Activities

Persona 1: Harold Johnson is a 73 year old father of 2 and grandfather of 4. He has been retired for 10 years and used to do real estate for a living. He got his undergraduate degree in finance from Rutgers and pursued a master's degree at Harvard. He loves to go out to eat with his wife and bird watching. Due to his real estate career, Harold is very wealthy and boasts ownership of 4 homes. He mainly presides in his largest house in Connecticut, where he lives with his wife. He uses his other homes mostly for storage or trips with his family.



Goals:

1. Find ideal hotels considering traffic, distance, food, weather...
2. Use Voice input rather than text input also Voice Output
3. No advertisements
4. Clear and concise web page
5. Enjoy luxury travel

User Scenario:

Harold has decided to search on google for a way to plan a vacation for him and his family. He saw our application in the search results and clicked it after checking out some of the high up results and finding that they did not satisfy his needs. He found that our application planned the trip mostly on its own with minimal input and decided to use our application for his needs. Upon entering the site, he clicked a button that said "Search Trip". The screen then brought him to a questionnaire in which he was sampled several questions by the system, such as "What are you looking to be booked?", "Where do you want to travel?", "How many people are going on the trip?". His responses were along the lines of "Complete vacation with a resort, transportation and food recommendations", "Florida", and "8", respectively. These questions would be laid out almost like a job application table, in which there are blank question fields on one page.

Persona 2: Mary Jane is a 21 year old college student with friends that all love to travel. She is about to graduate college at RPI and is a computer scientist that has a remote job as a software engineer lined up after graduation. She loves new technology and has been using AI a bunch just like the rest of her classmates. Her job will pay an annual salary of \$100,000. She is currently searching for a smaller home as she plans on traveling frequently and gets bored when staying in one place for too long. She enjoys surfing social media and meeting new people.



Goals:

1. Group of friends that want to travel minimally
2. Wants to simplify their trip and have a less stressful planning experience
3. Save money
4. Wants to spend more time having fun and less time worrying about her plans
5. Meet new people and want to find people who have the same interest travel together through our website

User Scenario:

After graduating Mary wants to take a grad-trip with her friend group as a celebration of getting through RPI. She remembers taking CSCI 4440 and remembers hearing about a group called Plotka that could help with travel needs. She searches for Plotka on firefox and see's it pop up in the top 10 searches. Upon entering the site, she clicked a button that said "Search Trip". The screen then brought her to a questionnaire in which she was sampled several questions by the

system, such as “What are you looking to be booked?”, “Where do you want to travel?”, “How many people are going on the trip?”, “When do you plan to take this trip?”, “What activities are you interested in?”. Her responses were “5 Star Hotel with pool”, “Europe”, and “6”, “July 17-28th, 2023”, and “Stargazing, photography, sightseeing, monuments, sculptures, art”, respectively. These questions would be laid out almost like a job application table, in which there are blank question fields on one page.

***Notes: This would be all we need for future reports - we do not need to repeat the basic stuff from report #1

Plotka Status Report #2

(Future Status Reports will be in this doc - most recent at the top of the document)

Status Paragraph:

Thus far, we have completed a basic web application, which can provide text answers to users. We created our vision statement, User Scenarios, Project Schedule, Github Repository, Plotka Discord Server, and basic project structure (frontend, backend, and web sockets). We also assigned work to everyone in the group so that we are aware of our responsibilities.

Accomplishments	Notes	Date
Vision Statement	Plotka vision statement	9/21
User Scenarios	Defined by Prof. Sturman	9/15 (in class)
Project Schedule	Plans for Project	9/21
Github created	GitHub Repository	9/12
Plotka Discord Server created	N/A	9/19
Work Assignment	Assigning works	9/19
The basic structure for our project	Including frontend, backend, and web socket	9/19

Plans (for next week):

- Design the API
- Design and draw a diagram for what the web page looks like
- Finish some basic AI API output on-site
- Plan to work on frontend after drawing how our web pages look like

Blockers:

- Schedule conflicts

*****Notes: This would be all we need for future reports - we do not need to repeat the basic stuff from report #1**

Plotka Status Report #1

(Future Status Reports will be in this doc - most recent at the top of the document)

Status:

We have created a basic website application which can provide text answers for users, but we fail to provide some other forms of answers like picture and mapping.

<https://gooeyai-avatar-one.glitch.me/>

Plans:

- Establish if our application is a website or mobile application
- Work on detailed vision statement
- Establish coding languages being used
- Establish coding standard
- Divide work amongst members/expectations
- Set milestones for the project/due dates
- List possible features and entitles for the project
- Figure out the structure of the project

Our application is a website application. If we have time, we will also create an app.

Coding language being used:

Frontend: Javascript

Backend: Python or Java

Coding standard:

- 1.add comments to explain what you write
- 2.Variable names cannot be generated casually
- 3.You can trade in space for time to improve efficiency

Works:

- 1.The provision of the basic navigation information
- 2.The creation of Navigation map
- 3.Arrangement of the elements in the main page
4. Try to get the locations of users by api

Plan:

Our plan is using five weeks to finish our project:

The first week:

Find all kinds of api which can help us to get data from other companies.

The second week:

Try to analyze and redesign data from api

The third week:

Try to know how to Display the information reasonably at the front end

The fourth week:

Try to create online api and connect backend and frontend

The final week:

Improve the performance of our application

The structure of project:

Home—>User input Page——> Guidance Page——>response page——> input page

Blockers:

- The size of backend project will lead to the failure of online upload of api.
- The hidden uncertainty for the response of application
- The slow speed of Information generation
- Most of us are unfamiliar with how to create an api and then put it online
- How to get the part of map from api and then provide tour guidance for the users
-

Work Completed:

- Last met on 9/11 (Monday)
- Set up team protocols and commitment
- Created discord group chat
- Planed small meeting group times

Contribution Summary Week 2

***Note:** Everyone worked on the Status Report and User Scenarios (this was in class)

Arnav Pawar:

- Created basic chatbot using an ai api.
- Worked on the prompt to get a desirable output.
- Looked and worked on a basic website and plan to get that fully working.
- Worked on vision statement(Specifically Features List)
- Worked on Project Schedule (planning things for each sprint)

Alex Litchfield:

- Vision Statement Sections:
 - Problem and Proposed Solutions
 - Feature List
 - Designed Logo
 - Overview of the Schedule
 - Niche
 - Overview of the development process:
- Project Schedule

Anson Decker:

- Worked on the Vision Statement
 - Elevator Pitch
 - Features
 - Competition
 - Stakeholders

Yuecheng Liu: Work on project schedule; work on Feature lists and Stakeholders parts of the vision statement; prepare to design the web page and to work on the frontend with Ziqun Wang.

Ziqun Wang: Build basic framework for the website including frontend, backend and websocket

--github

Prepare to design what the web page looks like with Yuecheng Liu

Worked on project schedule

Worked on vision statement

- feature list

Write simple template for api design --discord

Yihan Shao:

Provide a basic model include the basic frontend design and chatgpt

Be responsible for Plotka Status Report #1

Worked on vision statement:

- Value of efforts

- Risk of Product Success
- Competition